INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I , Fourth Semester Examination, February- 2025

Subject Code: BCAAE401

Subject Name: Software Engineering

Full Marks: 30 Set No: **1** Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What is the principal aim of the software engineering discipline? **2**

2. What do you understand by the term software life cycle? **2**

3. What is a prototype in prototyping model? **2**

4. What is meant by the 'size' of a software project? **2**

5. Write the advantages of using V-Model. **2**

6. What is evolution model? **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. What are the objectives of the feasibility study phase of software development? Explain the important activities that are carried out during the feasibility study phase of a software development project. **5**

8. Breifly explain different phases of prototyping model. **5**

9. Why spiral model is called a meta model? **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Explain the role of a software project manager. **10**

11. Explain different phases of classical waterfall model. **10**

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Subject Code:BCAAE401

Subject Name: Software Engineering

Full Marks: 30 Set No: 2 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Define the main objective of software testing? **2**

2. What are the characteristics of good software? **2**

3. What is the difference between software and a program? **2**

4. What is the purpose of Requirement Analysis? **2**

5. What are the major phases of the Software Development Life Cycle (SDLC)? **2**

6. What is the Spiral Model? **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Explain the Waterfall Model with its advantages and disadvantages. **5**

8. Describe the V-model of software development. **5**

9. Describe the RAD (Rapid Application Development) Model and its advantages. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Describe the Spiral Model with a diagram. How does it help in risk management? **10**

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11. Explain the Prototype Model in the software development life cycle. Discuss its advantages and limitations. **10**

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Subject Code: BCAAE401

Subject Name: Software Engineering

Full Marks: 30 Set No: 3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Distinguish between a program and a professionally developed software 2

2. What do you understand by control flow structure of a program? 2

3. What is a flow chart? 2

4. What do you understand by the term Software Development life cycle model (SDLC)? 2

5. Why is it necessary to model software life cycle and to document it? 2

6. Why the classical waterfall model is called an idealistic development model? 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Discuss the major advantages of the object-oriented design over the data flow oriented design method. 5

8. While using the iterative waterfall model to develop a commercial software for business application, discuss how the effort spent on the different phases is spread over time. 5

9. If the prototyping model is being used by a team developing a moderate sized software, is it necessary to develop an SRS document? Justify your answer. 5

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. a) Explain the important features of the agile software development model. 10
b) Compare the advantages and disadvantages of the agile model with iterative waterfall.

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c) A software development company is already experienced in developing a employee payroll software and has developed similar software for several customers (organizations). Assume that the software development company has received a request from a certain customer, which was still using manually processing of its employee payroll. For developing a payroll software for this organization which life cycle model amongst a) Iterative Waterfall model, b) Spiral life cycle model and , c) Agile software model, you suggest their best option to model the software. Justify your choice of the model.

11. a) Draw a schematic diagram to represent the iterative waterfall model of the software development. On your diagram represent the following: **10**
i) The phase entry and exit criteria for each phase.
ii) The deliverables that need to be produced at the end of each phase.

b) What are the objectives of feasibility study phase of software development? Explain the important activities that are carried out during the feasibility study phase of a software development project.

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